**Birla Institute of Technology & Science, Pilani**
**Work Integrated Learning Programmes Division**
**Second Semester 2023-2024**

**Mid-Semester Test**
**(EC-2 Regular – ANSWER KEY)**

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| Course No. | : DSECLZG529/AIMLCZG529 |
| Course Title | : Data Management for Machine Learning |
| Nature of Exam | : Open Book |
| Weightage | : 30% |
| Duration | : 2 Hours |
| Date of Exam | : ~~06/03/2021 or 19/03/2021 (FN/AN)~~ |

No. of Pages = 5
No. of Questions = 4

Note to Students:
1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q1. Imagine that you are building a next-generation telecom company, one that help users to avoid wastage of data from their existing data package subscription. First step is to deploy smart agent at customers' phone. These agents will report the application data use of the phone every hour, along with information on which applications are consuming the more data. Based on this, we would like to offer customers variable pricing in real time, based on when and how they run their favorite applications. **[1 + 2 + 1 + 1 + 2 = 7]**

a) Show a sample of the tuple generated out of the agent on phone.
b) Give two examples of exploratory analysis you will do on this data? Give details.
c) What can be the target variable in this example?
d) What sort of ML modelling technique can be utilized for it?
e) Which type of system architecture will be useful in this scenario? Draw a suitable architecture diagram for the same.

Answer:

a) {Timestamp, phone_no, {app1:data_usage, app2:data_usage, app3:data_usage,…}}
b) Each applications hourly / daily / day wise (week-day/week-end) data usage
   Day wise data requirements for a device
   Categorizing the data usages by types of applications – entertainment, work related
c) Recommended plan or subscription value
d) Classification – for plan recommendation , Regression – for plan value
e) Lambda – Both current and historical data taken into consideration for plan recommendation

Q2. Consider a data storage for University students. Each student data, stuData, which is in a file of size less than 64 MB (1 MB = $2^{20}$B). A data block (of default size 64MB) stores the full file data for a student of stuData_idN, where N =1 to 500. Assume that the cluster used to store these records consists of racks which has two DataNodes for processing each of 64 GB (1 GB = $2^{30}$B) memory. Cluster consists of 120 racks. Replication factor 3. **[1 + 3 + 2 + 1 + 1 = 8]**

a) How the files of each student will be distributed at a cluster?
b) How many students' data can be stored at one cluster?
c) What is the total memory capacity of the cluster (in TB) and DataNodes in each rack?
d) How many racks will be required if only 5000 student's data needs to be stored?
e) What will be the changes when a stuData file size <= 128 MB?

Answers:

a) How the files of each student will be distributed at a cluster?
Answer: Data block default size is 64 MB. Each students file size is less than 64 MB. Therefore for each student file one data block will suffice.

b) How many students' data can be stored at one cluster?
Answer:
A data block is in a DataNode. Each rack has two DataNodes each of memory size 64 GB.
Each node can store 64 GB / 64 MB = 1024 data blocks = 1024 student files
Each rack thus store 2 * 1024 data blocks = 2048 student files

Each block default replicates three times in DataNodes.
Number of students whose data can be stored in the cluster
= number of racks * number of files / 3 = 120 * 2028 / 3 = 81920

c) What is the total memory capacity of the cluster (in TB)?
Answer:
Total memory capacity of cluster
= 120 * 128 GB = 15360 GB
= 15 TB

d) How many racks will be required if only 5000 student's data needs to be stored?
Answer:
One rack will store 2048 students' data
For storing 5000 students data = 5000 / 2048 = 2.44 racks = 3 racks will be required (without replication)
With replication = 3 * 3 = 9 racks will be required

e) What will be the changes when a stuData file size <= 128 MB?
Answer: Changes will be that each node will have half number of data blocks.

Q3. An organization currently has an on-premise installation of an overall database system. This organization keenly looks into the developments in database systems, especially cloud databases. **[2 + 3 + 2 + 1 = 8]**

a) Name the type of cloud service that will match the requirement of hosting a database in the cloud. Enlist two cloud services from prominent vendors that can be helpful here.
b) Enlist three significant advantages the organization will achieve by migrating to the cloud.
c) List two changes that users of this system will face by migrating the databases to the cloud.
d) If the organization looking to support the analytical queries, what will be your suggestion for them?

Answer:

a) Database as a service - 1M
   AWS RDS, Azure SQL database – 1M
b) Three advantages by migrating to the cloud – Identifying each advantage 1 mark
   i. Elimination of physical infrastructure- In a cloud database environment, the cloud computing provider of servers, storage and other infrastructure is responsible for maintenance and keeping high availability. The organization that owns and operates the database is only responsible for supporting and maintaining the database software and its contents. In a DBaaS environment, the service provider is responsible for managing and operating the database software, leaving the DBaaS users responsible only for their own data.
   ii. Cost savings- Through the elimination of a physical infrastructure owned and operated by an IT department, significant savings can be achieved from reduced capital expenditures, less staff, decreased electrical and HVAC operating costs and a smaller amount of needed physical space.
   iii. DBaaS benefits also include instantaneous scalability, performance guarantees, failover support, declining pricing and specialized expertise.

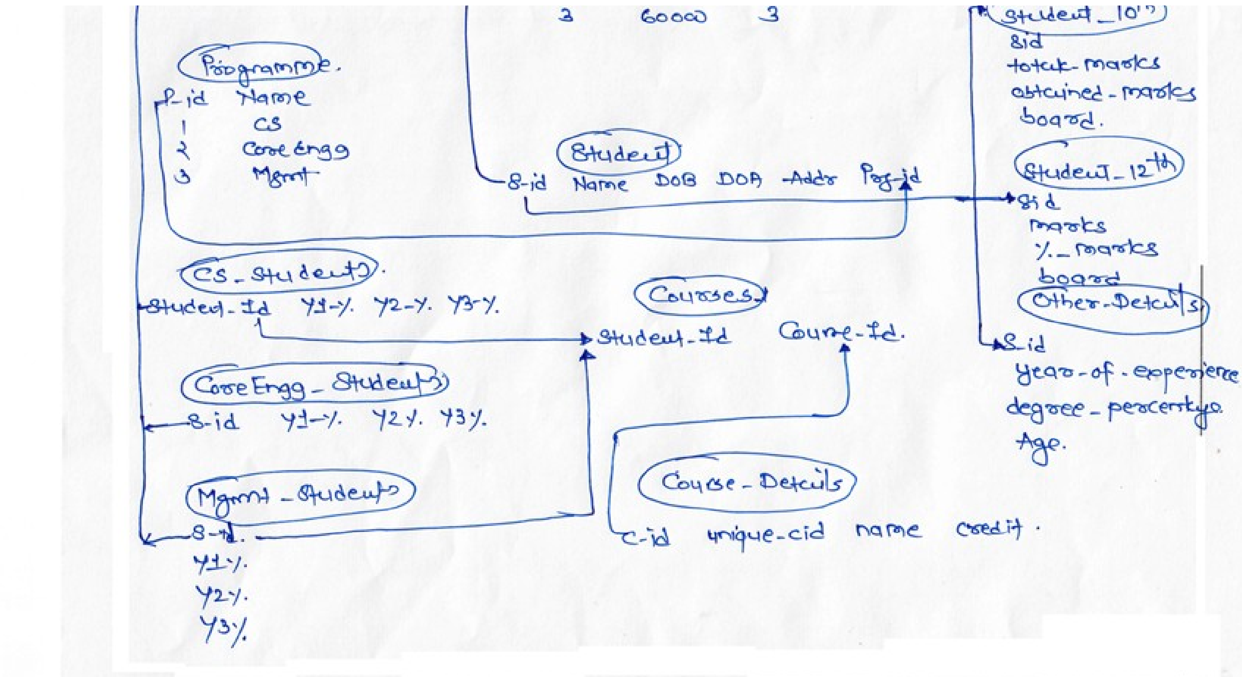
c) Two changes that users of this system will face by migrating the databases to the cloud – Identifying each change 1 mark
   i. Always internet connection will be required to access the database
   ii. Response time for queries might increase

d) Use any of cloud warehouse for data storage and analytics queries such as AWS Redshift, Google BigQuery, Snowflake etc.

Q4. The dean of placement division of Premiere Institute has collected data on their recent placement. To attract good students, it is important for the school to ensure that the students are placed with good salary package. The dean believed that the salary earned by a student at placement dependent on several variables.
The data collected by Dean has the following structure associated with it (look at next page). But he is not able to identify the significant attributes that are helping the students to earn good salaries. You are consulted by him to help in this exercise.

a) List down two major ambiguities which are present in the given ER diagram.
b) Flatten out the data set by applying the general data preprocessing techniques. Show a sample record of the same

c) Using the flattened out dataset, derive a model (just based on the intuition) that will predict what range of salaries can be earned by the student? **[2 + 4 + 1 = 7]**




Answer:

Answer:

a) Which student id needs to be considered? It's referred differently all over the places. What is the duration of the programmers? The year wise marks, degree marks are scattered.

Each ambiguity -1 mark, 1 * 2 = 2 marks

b) Flattened data set

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|---|---|---|---|---|---|---|---|---|---|
| Student_id | Name | Age | DOA | Addr | Programme | SSC marks | SSC Board | HSC marks | HSC board |
| Year_Of_Exp | Degree_perc | Y1 % | Y2 % | Y3 % | Y4% | Degree_years | Salary | | |

Removal of duplicate attributes – 1 mark

Removal of irrelevant attributes – 1 mark
Converting the attributes into appropriate forms – 1 mark
Sample record – 1 mark

c) Student can derive logistic regression model just taking two salary ranges into consideration like HIGH or LOW

Or they can think of building a decision tree where the outcome can be discrete salary ranges like LOW, AVERAGE, and HIGH etc.

Any one of the above model – 1 mark

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